Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. Which are characteristics of complex contagion? 1 / 1 point

- [x] Adoption depends on trusted sources and reinforcement.
  - Nice work
  - Feedback: Social reinforcement is key to complex contagion.
- [ ] A single exposure is enough for adoption.
- [x] Multiple exposures are often required.
  - Nice work
  - Feedback: This defines complex contagion.
- [ ] Influence flows randomly.

2. What concept does the Linear Threshold Model aim to represent more accurately? 1 / 1 point

- ( ) Disease spreads through physical contact.
- (•) Behavioral adoption through social reinforcement.
- ( ) Random walk processes in graphs.
- ( ) Network formation over time.
  - Nice work
  - Feedback: The model focuses on behaviors needing multiple exposures.

3. Which of the following are assumptions of the Independent Cascade Model? 1 / 1 point

- [ ] A node can infect neighbors multiple times.
- [x] Infection spreads probabilistically based on edge weights.
  - Nice work
  - Feedback: This is one of the key assumptions of the model.
- [x] Nodes attempt infection only once per neighbor.
  - Nice work
  - Feedback: This defines how infection attempts occur.
- [ ] The process always infects every node.

4. When does the cascade process stop in the Independent Cascade Model? 1 / 1 point

- ( ) After all nodes are infected.
- (•) When no new nodes are infected during a time step.
- ( ) When the last infected node is isolated.
- ( ) After three time steps.
  - Nice work
  - Feedback: This marks the termination condition of the cascade.

5. Why is the Linear Threshold Model deterministic? 1 / 1 point

- ( ) It uses a fixed sequence of time steps.
- ( ) It uses uniform probabilities.
- ( ) Each node has only one neighbor.
- (•) Outcomes are fixed if thresholds, weights, and seeds are fixed.
  - Nice work
  - Feedback: The model's determinism results from fixed parameters without randomness.

6. Why is the Independent Cascade Model considered more general than the SI model? 1 / 1 point

- (•) It allows each edge to have a different infection probability.
- ( ) It can model only directed networks.
- ( ) It runs faster computationally.
- ( ) It doesn't use a probabilistic framework.
  - Nice work
  - Feedback: This flexibility gives it more generality than the uniform SI model.

7. Which of the following influence adoption in the Linear Threshold Model? 1 / 1 point

- [x] Node's individual threshold.
  - Nice work
  - Feedback: Adoption depends on this threshold.
- [ ] Node's label in the graph.
- [ ] Number of steps elapsed.
- [x] Influence weights from neighboring adopters.
  - Nice work
  - Feedback: These determine the pressure to adopt.

8. What was the result of setting a 40% threshold in the cascade example? 1 / 1 point

- ( ) The behavior reached the same number of nodes as in the 30% scenario.
- (•) The cascade stopped after just a few adoptions.
- ( ) All nodes eventually adopted the behavior.
- ( ) Only one node adopted the behavior.
  - Nice work
  - Feedback: The higher threshold made it harder to sustain the cascade, leading to early termination.

9. What changes are part of the simplified linear threshold model? 1 / 1 point

- [x] Assuming all neighbors have equal influence.
  - Nice work
  - Feedback: This is a defining simplification of the model.
- [ ] Introducing randomness in edge influence.
- [ ] Assigning weights based on social status.
- [x] Using a fixed fraction of neighbors required for adoption.
  - Nice work
  - Feedback: This simplifies threshold assignment across the network.

10. Which enhancements could make the Independent Cascade Model more flexible? 1 / 1 point

- [x] Letting infection probability decrease over time.
  - Nice work
  - Feedback: This allows for time-varying dynamics, adding realism.
- [x] Allowing nodes to infect neighbors over multiple steps.
  - Nice work
  - Feedback: This introduces extended infectious periods.
- [ ] Fixing the probability of infection for all edges.
- [ ] Adding thresholds to each node.

11. In the Independent Cascade Model, what happens when a node becomes infected? 1 / 1 point

- ( ) It infects all its neighbors with certainty.
- ( ) It attempts to infect neighbors repeatedly across time steps.
- ( ) It chooses one neighbor at random to infect.
- (•) It gets only one chance to infect each of its susceptible neighbors.
  - Nice work
  - Feedback: This rule defines the key mechanism of the model.

12. What distinguishes the Independent Cascade Model from the SI model? 1 / 1 point

- (•) Edges can have unique transmission probabilities.
- ( ) All nodes are infected simultaneously.
- ( ) Infected nodes can never infect others.
- ( ) The model removes infected nodes immediately.
  - Nice work
  - Feedback: This is a core feature of the Independent Cascade Model, unlike SI, which assumes uniform probabilities.

13. What determines whether a node adopts a behavior in the Linear Threshold Model? 1 / 1 point

- ( ) A global infection parameter.
- ( ) Whether it has the most connections.
- ( ) The number of seed nodes in the network.
- (•) Whether the sum of influence weights from neighbors exceeds its threshold.
  - Nice work
  - Feedback: This is the defining rule for adoption in the model.

14. What must be true about edge weights in the Linear Threshold Model? 1 / 1 point

- ( ) All edges have the same weight.
- ( ) Edges must be bidirectional.
- (•) The sum of incoming weights for a node is at most one.
- ( ) They must sum to more than one.
  - Nice work
  - Feedback: This ensures influence is normalized and manageable.

15. Which node would adopt a behavior sooner in the Linear Threshold Model? 1 / 1 point

- ( ) A node already adopted by others.
- (•) A node with a low threshold and influential neighbors.
- ( ) A node with no incoming edges.
- ( ) A node with a high threshold and few neighbors.
  - Nice work
  - Feedback: This combination leads to earlier adoption.

16. In the simplified linear threshold model, what does each edge weight represent? 1 / 1 point

- ( ) Randomly assigned influence from the network generator.
- ( ) The actual probability of adoption.
- ( ) Influence based on distance in the graph.
- (•) Equal influence from each neighbor based on node degree.
  - Nice work
  - Feedback: In the simplified model, each neighbor's influence is 1 divided by the node's degree.

17. Which of the following statements about the simplified linear threshold model are true? 1 / 1 point

- [ ] All nodes will eventually adopt the behavior.
- [ ] Nodes adopt the behavior if one neighbor adopts.
- [x] Adoption depends on the fraction of neighbors who have adopted.
  - Nice work
  - Feedback: Fractional influence is central to this simplified model.
- [x] Each edge to a node has the same weight.
  - Nice work
  - Feedback: Equal edge weight is a key simplifying assumption.

18. What determines the infection probability in a directed Independent Cascade network? 1 / 1 point

- ( ) Node color.
- ( ) Number of infected neighbors.
- (•) Edge-specific transmission probabilities $P_{u,v}$.
- ( ) Threshold values of nodes.
  - Nice work
  - Feedback: The directed edge probabilities define infection likelihoods between nodes.